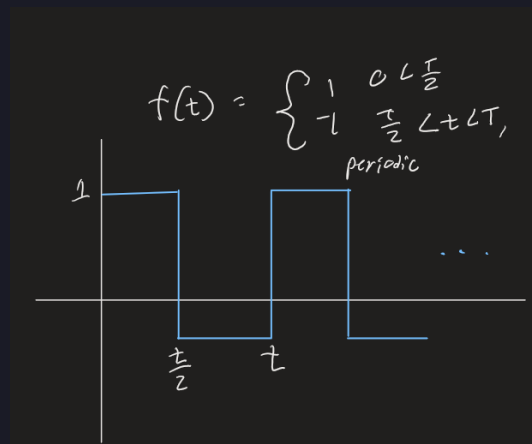


Fourier pt 2

What if we want to represent a periodic function instead?

We could use a sum of periodic functions to make the true function. Instead of expanding a series with x^n s., could we use $A \sin(kx)$ s?

Lets imagine we want to represent a square wave?



We want to make the period T so that it fits with this square wave.

We can make an initial guess:

$$f(t) = a_0 + \sum_{n=1}^{\infty} a_n \cos\left(\frac{2\pi n t}{T}\right) + b_n \sin\left(\frac{2\pi n t}{T}\right)$$

This looks like a vector space - we have a linear combination of two basis.

It would be nice if these were orthogonal, so we define an inner product that would make these two yield zero.

If our function has a non zero average, then we know that has to come from a_0 , since that is the only function with a non zero average in our guess.

We can write out:

$$\int_0^T \cos\left(\frac{2\pi n t}{T}\right) \cos\left(\frac{2\pi m t}{T}\right) dt = \begin{cases} 0 & n \neq m \\ -\frac{T}{2} & n = m \neq 0 \\ T & n = m = 0 \end{cases} \quad (1)$$

only when $m = n$ we will get something that doesn't average to zero over a full period. All the others oscillate around zero.

We know that we can represent

$$\cos(\omega t) = \frac{e^{i\omega t} + e^{-i\omega t}}{2} \quad (2)$$

$$\sin(\omega t) = \frac{e^{i\omega t} - e^{-i\omega t}}{2i} \quad (3)$$

Consider the following integral:

$$\int_0^T e^{in\omega t} e^{-im\omega t} dt, \quad \omega = \frac{2\pi}{T} \quad (4)$$

$$= \int_0^T e^{i(n-m)\omega t} dt \quad (5)$$

$$= \left. \frac{e^{i(n-m)\omega t}}{i(n-m)\omega} \right|_0^T \quad (6)$$

$$(7)$$

If $n = m$, this explodes to infinity, so we would have to go back and integrate $\int_0^T 1$. Otherwise, we have this thing.

$$\frac{e^{i(n-m)\omega T} - 1}{i(n-m)\omega} \quad (8)$$

$$= \frac{e^{i(n-m)2\pi} - 1}{i(n-m)\omega} = 0 \quad (9)$$

Lets use what we just developed and go back to

$$\int_0^T \cos(n\omega t) \cos(m\omega t) dt \quad (10)$$

$$= \int_0^T \left(\frac{e^{in\omega t} + e^{-in\omega t}}{2} \right) \left(\frac{e^{im\omega t} + e^{-im\omega t}}{2} \right) dt \quad (11)$$

We get

$$\frac{1}{4} \int_0^T \underbrace{e^{i(n+m)\omega t}}_0 + \underbrace{e^{-i(n+m)\omega t}}_0 + \underbrace{e^{i(n-m)\omega t}}_{T\delta_{nm}} + \underbrace{e^{-i(n-m)\omega t}}_{T\delta_{nm}} dt \quad (12)$$

We know that were whipping around the unit circle some positive number of times, so the first two e's are 0. The next two are zero unless $n=m$, in which we get T .

This gets us

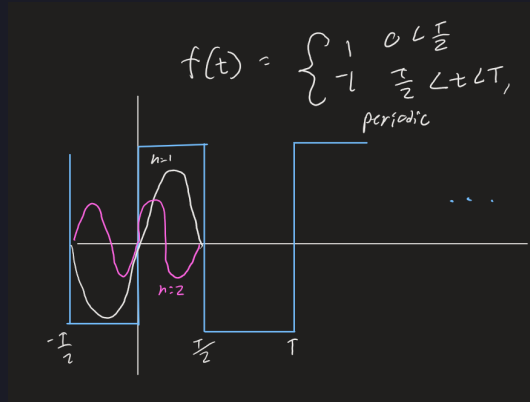
$$\frac{T}{2} \delta_{nm} \quad (13)$$

So we have our basis functions that are orthogonal if we let the inner product be the integral over the period of the product of the vectors. Its zero everywhere except where $n = m$.

If we shift our frame to only look at the square wave from $-\frac{T}{2}$ to $\frac{T}{2}$, we see that the function looks odd.

We know that our $a_n \cos(n\omega t)$ has $a_n = 0$,

$$f(t) = -f(-t), \text{ and } \cos \text{ is even} \quad (14)$$



If we have even n 's, then we have equal contribution to the positive and negative regions, so they would integrate to 0. We can therefore only have odd integers that we can fit.

We multiply $f(t) = \dots$ by $\sin m\omega t$ and integrate from $-\frac{T}{2}$ to $\frac{T}{2}$
We get

$$\int_{-\frac{T}{2}}^{\frac{T}{2}} f(t) \sin(m\omega t) dt \quad (15)$$

$$= \int_{-\frac{T}{2}}^{\frac{T}{2}} \left[a_0 + \sum_{n=1}^{\infty} a_n \cos n\omega t + b_n \sin(n\omega t) \right] \sin m\omega t dt \quad (16)$$

We can reduce this with what we learned (0 mean, so no a_0 , no cosines, and only odd sines)

$$\int_{-\frac{T}{2}}^{\frac{T}{2}} b_n \underbrace{\sin(n\omega t) \sin(m\omega t)}_{\delta_{nm} \frac{T}{2}} dt \quad (17)$$

We therefore have

$$I_{nm} \equiv 2 \int_0^{\frac{T}{2}} \sin(m\omega t) dt \quad (18)$$

$$= 2 - \frac{\cos(m\omega t)}{m\omega} \Big|_0^{\frac{T}{2}} \quad (19)$$

$$= -\frac{2}{m\omega} \left[\cos\left(\frac{m\omega T}{2}\right) - 1 \right] \quad (20)$$

$$= -\frac{2}{m\omega} [\cos(m\pi) - 1] \quad (21)$$

$$= \frac{4}{m\omega} \text{ if } m \text{ is odd} \quad (22)$$

$$= \frac{b_m T}{2} \quad n=m \quad (23)$$

and we have nothing if m is even.

We therefore get

$$b_m = \begin{cases} \frac{2}{T} \frac{4}{m\omega} & m \text{ odd} \\ 0 & m \text{ even} \end{cases} \quad (24)$$

$$b_m = \frac{4}{m\pi} \quad m \text{ odd} \quad (25)$$

$$f(t) = \sum_{n \text{ odd}} \frac{4}{n\pi} \sin\left(\frac{2\pi n t}{T}\right) \quad (26)$$